

Laboratory Session #3

Deliverables

This assignment is to be completed in pairs and will be submitted to Crowdmark (PDF). Pairing allocations are posted in the lab session and can also be found on LEARN.

In this assignment, you will:

- Decipher code written by others
- Develop helpful input and output statements
- Insert concise, descriptive comments
- Create arrays, use basic array functions, and perform mathematical operations with arrays

Your Crowdmark submission should contain:

1. A header comment block, formatted as follows:
% Authors: Names of those working on the assignments
% Purpose: Short description of what the code does
% Date Created: Today's date
% Acknowledgements: Any outside sources used to complete
% assignment, including website, books, and people (other
% than the teaching team)
2. The output of the original code for each test case
3. The code that you wrote and its output for each test case
4. Your answers to the questions included in the assignment.

Crowdmark Submission with Partners

Prior to submitting your assignment work to Crowdmark, you need to tell Crowdmark that you are part of a group. To do this, complete the following steps:

- Navigate to the Lab Assignment submission area in Crowdmark.
- Click the "Add Group Members" button at Crowdmark and choose your group for this assignment. You can also wait for someone else to add you to their group. *Note that after the assignment has been submitted, your group cannot be changed.*
- Each group member you add will receive an email notification and shared access to this page.
- Any group member may submit the assignment or edit group members.

Problem Description

Prof. Smith's concrete class just completed their lab on concrete mix design. For this lab, each group was assigned a different mix design number and was responsible for mixing and casting a concrete test cylinder based on the ingredients for that mix design (e.g., different sizes of aggregate, differing amounts of cement, different additives, different water ratios). The students then tested the compressive strength of their concrete cylinders. The results of these tests for Prof. Smith's three lab sections are shown below.

Mix Design Number	Compressive Strength [psi]		
	Lab 1	Lab 2	Lab 3
1	5242	3357	3460
2	2537	2280	2323
3	3152	6582	3409
4	3085	2935	3339
5	3010	2706	2890
6	2163	2614	2631
7	2599	2715	1574
8	3338	3311	2892
9	1685	2537	2716
10	3105	3436	3384
11	7522	7607	7956
12	2910	2960	2440

Let's write a code to help Prof. Smith's students analyze the results of their laboratory testing. Begin by entering the data from the table above as a single 12×4 matrix in your MatLab script file. The first column should contain the mix design numbers, and columns 2–4 should contain the compressive strength results for Lab 1, Lab 2, and Lab 3, respectively.

Part A

The students from each lab session would like to get an overall picture of how their specimens performed relative to each other. Using the matrix you created from the table above, write code to:

- Determine the minimum, average, standard deviation, and maximum compressive strength values for each lab session. You should calculate these statistics using the Lab 1, Lab 2, and Lab 3 columns of the matrix. You should end up with 3 values for each statistic.
- Report the information from (i) to the class in an informative tabular format, similar to the following:

Lab	Weakest	Avg. Strength	Std. Dev.	Strongest
---	-----	-----	-----	-----
1	1685	3362	1563	7522
2	2280	3587	1689	7607
3	1574	3251	1581	7956

Remember that additional contextual information can be added before and after your tables to convey other important information to your users (e.g., units).

Part B

Knowing that standard concrete mixes typically have a range of compressive strengths between 2,500 – 4,000 psi, Prof. Smith's students are concerned about some of the results they're seeing in their experiments. Adding to your script file from Part A, write a code that helps the students remove these apparent outliers by:

- i. Splitting the original 12x4 matrix into 3 12x1 vectors, which hold the results from each lab session.
- ii. Identifying and reporting the mix design numbers of all outliers for each lab session, where an outlier is considered to be a compressive strength less than 2,500 psi or greater than 4,000 psi. As an example of how these values might be reported:

The mix designs which produced outliers for Lab 1 include:
 Mix 1 Mix 6 Mix 9 Mix 11

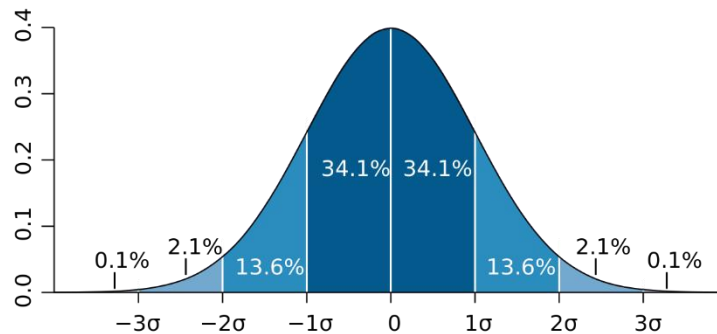
Hint: The indices of vector elements meeting a certain condition can be found using the `find()` function.

- iii. Removing the outliers from your vectors and recalculating the statistics on the modified lab results. Report those statistics to the class in a format similar to your outputs from Part A.

Extra Reference Material

The standard deviation of a set of data is a measure of how spread out your data set is. If the standard deviation is small, the data is tightly grouped together. If the standard deviation is large, the data is spread out.

The standard deviation also helps us describe the steps we can take away from our mean value while still finding data that has probably come from the same sample set. The image below shows that by taking a single standard deviation step to either side of the mean, we encapsulate ~70% of the data from our sample set.

**What you need to submit**

To Crowdmark:

- Your code and the resulting command window display.